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# Think About the Reader Once, Not Every Sentence: Per-Sentence Audience Simulation in Chain-of-Thought Hurts LLM Writing

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## Abstract

Large language models often write in a fluent-but-generic voice, as if pitched at no one in particular. A natural explanation is that they model the reader too weakly, and a natural fix is to force the model to simulate the reader inside its own chain-of-thought—roughly once per sentence—so that every next sentence is chosen against a fresh prediction of how the reader reacts. We give this hypothesis its first direct test. We generate 60 audience-targeted writing tasks under four conditions—plain generation, generic step-by-step reasoning (a token-matched control), one-shot audience modeling (SIMTOM), and the proposed per-sentence audience simulation—and evaluate them with two separate, cross-family judge models: one role-plays the target reader to rate audience fit, the other scores overall writing quality via blind, order-balanced pairwise comparison. Across paired significance tests, effect sizes, and Holm correction, the specific hypothesis is refuted while its underlying intuition is vindicated. Modeling the reader does help: a single rich audience model written once up front is the best condition on every metric—audience fit, engagement, genuine audience adaptation, and prose diversity—at lower cost. Per-sentence simulation is the worst or near-worst condition: it loses pairwise on overall quality to all three baselines (win-rates 0.08–0.25, all  $p < 0.001$ ), adapts *less* to the reader rather than more, does not reduce homogenization, and is the longest and most token-expensive. The result replicates across two generator families. The lesson is about granularity: think about the reader once, deeply—not reactively, sentence by sentence.

## 1 Introduction

Large language models write fluently, yet their prose is often flat and generic—polished sentences that read as if written for no one in particular. A common diagnosis is that the model fails to model its reader: it produces text that is locally correct but pitched at a default, average audience rather than the specific person who will read it. If this diagnosis is right, there is an appealing, cheap fix. Rather than fine-tune the model, we could simply ask it, in its chain-of-thought, to *simulate the reader*—and to do so continuously, roughly once per sentence, so that each next sentence is chosen against a fresh prediction of how the reader is reacting: confused, bored, skeptical, or curious. This is an inference-time, prompt-only intervention that anyone could apply. *Does it work?*

**Why this matters.** Writing quality is a pervasive, high-value problem for LLMs used as assistants, tutors, and technical communicators. If verbalizing an incremental, per-sentence audience simulation genuinely improved writing, it would be an unusually cheap win: no training, no reward model, no external listener—just a better prompt. The intuition also has strong roots. The pragmatics literature shows that speakers who model their listener communicate more successfully [Andreas and Klein, 2016, Fried et al., 2023, Takmaz et al., 2023], and the Theory of Mind (ToM) literature shows that default LLM social reasoning is brittle [Ullman, 2023, Shapira et al., 2023], suggesting that an explicit modeling step might help.

**The gap.** Despite this, the exact intervention has never been tested for writing. Every prior audience-aware method places the audience model *outside* the generator’s own reasoning: a decode-time gradient [Takmaz et al., 2023], a rerank by a separately trained listener [Andreas and Klein, 2016, Anonymous, 2025b], an external reward [Anonymous, 2023], or a formal belief update [Anonymous, 2025c]. Per-sentence mental-state simulation *inside* natural-language chain-of-thought exists only for comprehension QA and dialogue [Anonymous, 2024, 2025d,e], never for open-ended writing quality. Meanwhile, Chakrabarty et al. [2025] show that generic reasoning does *not* improve writing quality—so a gain from an *audience-specific* chain-of-thought would be a genuinely non-trivial result, and any claimed gain must be isolated from “more thinking tokens.”

**Our approach.** We run the first direct test of the intervention (section 3). We build 60 audience-targeted writing tasks—explaining a concept to an 8-year-old, a non-technical executive, a PhD expert, or a skeptic, plus creative stories with an assigned readership—and generate each under four conditions that differ *only* in how the model reasons about the reader: PLAIN (write directly), GENERIC-COT (generic “think step by step,” a token-matched control), SIMTOM (describe the reader once, then write), and PER-SENTENCE (the intervention: plan sentence by sentence, and after each sentence simulate how this reader reacts, then choose the next). Every condition is told exactly who the reader is, so any difference is attributable to the *reasoning strategy*, not to privileged audience information. We judge only clean final text with two separate, cross-family models—one role-playing the reader for audience fit, one scoring overall quality by blind pairwise comparison—and add an adaptation-gap control that measures whether a method *actually uses* the audience rather than just writing well in general.

**What we find.** The specific hypothesis is refuted, but its underlying intuition is vindicated. Audience modeling does help—but a single, rich audience model written once up front (SIMTOM) is the clear winner on every metric, at lower cost. The proposed per-sentence simulation is the worst or near-worst condition: it loses pairwise on overall quality to *all three* baselines (win-rates 0.083–0.250, all  $p < 0.001$  after Holm correction), scores lowest on engagement and audience fit, shows *no* better audience adaptation (in fact the smallest adaptation gap, 6.04 vs. SIMTOM’s 7.00), does not reduce homogenization, and is the longest (170 words vs. 147) and most token-expensive condition. The ordering replicates with a completely different generator (LLAMA-3.3-70B). The takeaway is blunt: “*think about the reader*” *helps* writing; “*re-simulate the reader every sentence*” *hurts* it.

### Contributions.

- We give the first direct, controlled test of per-sentence audience simulation in chain-of-thought for open-ended writing, isolating it from a token-matched generic-reasoning control and from privileged audience information.
- We show the intervention *backfires*: it loses on overall quality to plain generation, generic reasoning, and one-shot audience modeling, and—via an adaptation-gap control—adapts to the reader *less*, not more, than a one-shot audience model.
- We show the intuition behind it is nonetheless correct: one-shot audience modeling (SIMTOM) is the best condition on audience fit, engagement, adaptation, and diversity, at lower token cost, identifying *granularity* as the key design axis.
- We replicate the full ordering across two generator families and release all 3,876 cached generations and judgments for reproducibility.

**Organization.** section 2 positions our work in pragmatics, ToM, and writing-quality research. section 3 describes the tasks, conditions, models, and metrics. section 4 reports the main comparison, the adaptation-gap control, homogenization and cost, and the cross-family replication. section 5 interprets *why* per-sentence simulation hurts and discusses limitations, and section 6 concludes.

## 2 Related Work

**Audience-aware generation and pragmatics.** The Rational Speech Acts tradition models a speaker who chooses each utterance by simulating how a listener will interpret it relative to alternatives [Andreas and Klein, 2016, Fried et al., 2023]. Andreas and Klein [2016] sample candidate descriptions and rerank them with a simulated listener, gaining accuracy while a small fluency weight keeps language natural; notably, the reranking reasoning did not distill into a feedforward network, suggesting explicit inference-time reasoning matters. Takmaz et al. [2023] steer a frozen speaker at decode time with a listener simulator, and find that modeling a *specific* reader (audience-aware) beats modeling

a generic one (self-aware), which beats no modeling at all. Anonymous [2025b] reverse-generate what an audience thinks a program does and rerank for distinctiveness, and Anonymous [2025c] maintain a formal per-turn belief over a listener’s private state, gaining most when that belief is genuinely uncertain. Anonymous [2023] treat audience preference as an external reward. Across all of these, audience modeling lives *outside* the generator’s own reasoning—as a gradient, a reranker, a reward, or a belief-update module. Unlike these methods, we place audience modeling *inside* the generator’s natural-language chain-of-thought, and ask whether doing so per-sentence improves open-ended prose.

**Per-sentence and reflective mental-state simulation.** A separate line applies incremental mental-state reasoning inside the model’s own text. Anonymous [2024] prompt the model, per sentence, for who perceived each piece of content, filter to the target’s perspective, and then answer, yielding large false-belief gains. Anonymous [2025d] trace mental states with an incremental, hypothesis-driven agent, and Anonymous [2025e] infer a partner’s belief-desire-intention, predict their reply, and reflect. These are the closest method analogs to our intervention, but all target *comprehension QA or dialogue*, where the goal is to answer a question about a mind, not to *write well*. We port the per-sentence template to writing and find that what helps comprehension hurts prose.

**Theory of Mind in LLMs, and the chain-of-thought caution.** Default LLM ToM is real but brittle, collapsing under trivial adversarial perturbations [Ullman, 2023, Shapira et al., 2023]. Crucially, Shapira et al. [2023] warn that chain-of-thought can inflate ToM scores through leaked task structure and shortcuts, so a positive result from an audience-simulation prompt must be checked against controls and a separate judge rather than taken at face value. Perspective-taking prompts such as SIMTOM already help ToM QA [Anonymous, 2025f], and pragmatic competence grows across training stages [Anonymous, 2025a]. We heed the caution directly: we judge only clean extracted text with judges in different families from the generator, and add an adaptation-gap control that rules out a generic-quality confound.

**Writing quality and homogenization.** LLM prose is fluent but homogenized, and strong judges struggle to separate good writing from bad [Anonymous, 2026]. Most relevant to us, Chakrabarty et al. [2025] show that generic chain-of-thought and reasoning models do *not* improve writing quality, while span-level self-editing does. This makes our question sharp: if audience-specific reasoning helped writing, it would be a non-trivial exception to the “more reasoning does not help writing” finding. Building on their testbed and the homogenization metrics of Anonymous [2026], we measure quality, audience fit, adaptation, diversity, and cost together. Our result extends theirs: even audience-specific reasoning fails to help when applied at fine (per-sentence) granularity, whereas a coarse one-shot audience model does help.

### 3 Methodology

**Research question and hypotheses.** We ask whether having an LLM simulate the target reader in its chain-of-thought roughly once per sentence produces writing that is (i) better fit to that reader, (ii) higher in overall quality, and (iii) more genuinely adapted to the reader, compared to plain generation, a token-matched generic reasoning control, and one-shot audience modeling. We decompose this into five pre-registered hypotheses: **H1** (audience fit): PER-SENTENCE beats all baselines on a separate reader model’s fit rating; **H2** (quality): PER-SENTENCE beats baselines on a cross-family quality judge; **H3** (adaptation): PER-SENTENCE shows the largest audience-adaptation gap; **H4** (not just tokens): PER-SENTENCE beats generic reasoning despite both adding thinking tokens; **H5** (diversity/cost): PER-SENTENCE reduces generic sameness at acceptable cost. The independent variable is the generation *condition*; dependent variables are audience-fit score, pairwise quality win-rate, adaptation gap, diversity metrics, and length/token cost.

#### 3.1 Task set

We use **audience-targeted writing**, the sharpest test of audience modeling, comprising 60 tasks. **48 explanatory** tasks pair a concept with a specific reader: “Explain {concept} for {audience}” in roughly 150 words. Twelve concepts span science, finance, and technology (vaccines, compound interest, black holes, how an LLM generates text, antibiotic resistance, blockchain, seasons, mRNA, the stock market, encryption, photosynthesis, greenhouse warming), crossed with four audiences that vary in expertise and disposition: **an 8-year-old child, a non-technical executive, a PhD-level**

**expert**, and a **skeptic who distrusts the topic**. 12 creative tasks draw prompts from WRITING-PROMPTS [Fan et al., 2018] with an assigned target readership, for breadth.

### 3.2 Conditions

The only manipulation is *how the model reasons about the reader*; crucially, every condition is given identical information about who the reader is, so any difference is attributable to the reasoning strategy rather than to privileged audience information. Generation uses a clean two-step protocol—a reasoning turn, then a “write ONLY the final piece” turn—so the judged text never contains chain-of-thought scaffolding.

- **PLAIN**: no reasoning turn—write directly (the “annoying” egocentric default).
- **GENERIC-COT**: “think step by step to make it excellent” (generic reasoning; **token-matched control** that isolates audience simulation from added thinking tokens).
- **SIMTOM**: describe the reader once (background, knowledge, what they care about, tone), then write.
- **PER-SENTENCE (intervention)**: plan sentence by sentence; after each sentence, simulate how *this reader* reacts (confused, bored, skeptical, curious), then choose the next sentence.

### 3.3 Models

We deliberately keep the generator and both judges in **different model families** to avoid self-preference and shortcut inflation [Shapira et al., 2023]. The **generator** is GPT-4.1-MINI (primary) with LLAMA-3.3-70B for replication. The **quality judge** is CLAUDE-SONNET-4.5, used for blind, order-balanced pairwise comparison. The **audience/listener model** is GEMINI-2.5-FLASH, which *role-plays the target reader* to give absolute fit, understanding, and engagement ratings (1–10) and pairwise “which serves this reader better” judgments, following communication-based evaluation [Fried et al., 2023]. All calls run through OpenRouter with a fixed seed of 42.

### 3.4 Evaluation metrics

1. **Audience fit / understanding / engagement**: the audience model, role-playing the reader, rates each piece 1–10.
2. **Overall writing quality**: the cross-family judge compares pairs blind, in both orders, controlling for position bias; we report win-rate (0.5 = tie).
3. **Adaptation gap** (the H3 control): each concept is written for a *child* and for an *expert*; both readers rate both pieces. The gap is matched – crossed fit. A method that truly tailors to the reader scores high for its intended reader and low for the other, yielding a large gap; a generic-quality method yields a gap near zero. This isolates real audience use from a generic quality boost.
4. **Homogenization**: distinct-2 and self-BLEU computed within each condition.
5. **Cost**: output length in words and reasoning (“thinking”) characters.

### 3.5 Protocol and reproducibility

Generation uses temperature 0.7; judging uses temperature 0.0; the seed is 42 throughout. Every model call is disk-cached by a hash of (model, messages, params), so the pipeline is fully reproducible and re-runs are free. In total we ran 240 primary generations, 192 replication generations, and roughly 1,400 judge calls, yielding 3,876 cached responses at an estimated API cost of \$8–15. For statistics we use paired designs across tasks: Wilcoxon signed-rank tests on paired differences, bootstrap 95% confidence intervals for win-rates (against a null of 0.5) and mean differences, Cohen’s *d* / rank-biserial effect sizes, and Holm–Bonferroni correction [Holm, 1979] across the multiple condition comparisons, at  $\alpha = 0.05$ . An initial marker-based text extractor leaked audience-analysis preamble into about 23% of SIMTOM pieces; we replaced it with the two-step protocol (zero leaks) and re-ran everything, so all reported numbers come from the clean pipeline.

Table 1: Absolute audience ratings (GEMINI-2.5-FLASH role-playing the reader, 1–10;  $n = 60$ ; higher is better). PER-SENTENCE, the intervention, is near the bottom on every dimension—below both reasoning baselines and only marginally above PLAIN. SIMTOM is best throughout. Best results in **bold**.

| Metric        | PLAIN | GENERIC-CoT | SIMTOM      | PER-SENTENCE ( <i>ours</i> ) |
|---------------|-------|-------------|-------------|------------------------------|
| Audience-fit  | 6.37  | 7.12        | <b>7.28</b> | 6.52                         |
| Engagement    | 5.80  | 6.45        | <b>6.62</b> | 5.93                         |
| Understanding | 9.15  | 9.30        | <b>9.32</b> | 9.08                         |

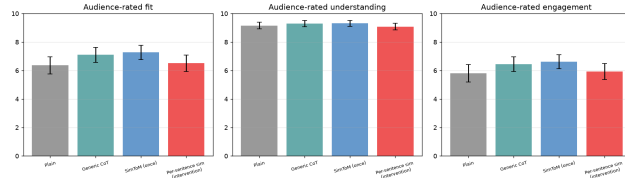


Figure 1: Absolute audience ratings with 95% confidence intervals. The intervention (PER-SENTENCE) sits near PLAIN and below both GENERIC-CoT and SIMTOM on audience fit, engagement, and understanding; the one-shot audience model SIMTOM is highest on all three.

## 4 Results

We organize results around the five hypotheses. Across the board, the intervention fails: it is near the bottom on absolute ratings (section 4.1), loses every pairwise comparison on overall quality (section 4.2), adapts to the reader *least* (section 4.3), and is the most homogenized and most expensive condition (section 4.4). The ordering replicates with a second generator (section 4.5). Throughout, SIMTOM—a one-shot audience model—is the best condition.

### 4.1 Absolute audience ratings (H1)

Table 1 reports the audience model’s ratings while role-playing the reader. PER-SENTENCE is near the bottom on every dimension, barely above PLAIN and clearly below both GENERIC-CoT and SIMTOM. Paired Wilcoxon tests (intervention vs. each baseline), Holm-corrected, show PER-SENTENCE is significantly *worse* than SIMTOM on fit ( $p = 0.0001$ ), engagement ( $p = 0.0002$ ), and understanding ( $p = 0.057$ ), and worse than GENERIC-CoT on fit ( $p = 0.0007$ ) and engagement ( $p = 0.0025$ ); it is statistically indistinguishable from PLAIN. The proposed intervention thus fails to improve audience fit—**H1 is refuted** (figure 1).

### 4.2 Pairwise quality and fit (H2, H4)

Table 2 reports blind, order-balanced pairwise win-rates for the intervention against each baseline. On **overall quality** the intervention loses to all three baselines—most heavily to SIMTOM, winning only 8.3% of comparisons—and on **audience fit** it loses to GENERIC-CoT and SIMTOM while merely tying PLAIN. All effects survive Holm correction. Because PER-SENTENCE loses to the token-matched GENERIC-CoT control despite both adding reasoning tokens, the failure is not explained by “more thinking”—**H2 and H4 are both refuted** (figure 2).

### 4.3 Does the method actually use the audience? (H3)

If per-sentence simulation genuinely tailored writing to the reader, it should show the *largest* adaptation gap. Instead it shows the *smallest*, tied with PLAIN (Table 3). Its gap of 6.04 is significantly smaller than SIMTOM’s 7.00 ( $\Delta = -0.96$ , Cohen’s  $d = -1.27$ ,  $p = 0.004$  Holm-corrected) and smaller than GENERIC-CoT’s 6.67 ( $\Delta = -0.62$ ,  $d = -0.75$ ). The one-shot audience model, which commits to a single coherent stance about the reader, adapts *most*. **H3 is refuted**: the intervention does not use the audience more—it uses it less (figure 3).

Table 2: Pairwise win-rate of PER-SENTENCE vs. each baseline ( $n = 60$ ; both orders averaged; 0.5 = tie; lower means the intervention loses more often). The intervention loses on overall quality to *all three* baselines and on audience fit to the two stronger baselines. All significant win-rates in **bold**.

| Comparison                   | Overall quality (CLAUDE-SONNET-4.5) | Audience-fit (GEMINI-2.5-FLASH) |
|------------------------------|-------------------------------------|---------------------------------|
| PER-SENTENCE vs. PLAIN       | <b>0.250</b> ( $p < 0.001$ )        | 0.475 ( $p = 0.67$ , tie)       |
| PER-SENTENCE vs. GENERIC-COT | <b>0.150</b> ( $p < 0.001$ )        | <b>0.250</b> ( $p < 0.001$ )    |
| PER-SENTENCE vs. SIMTOM      | <b>0.083</b> ( $p < 0.001$ )        | <b>0.200</b> ( $p < 0.001$ )    |

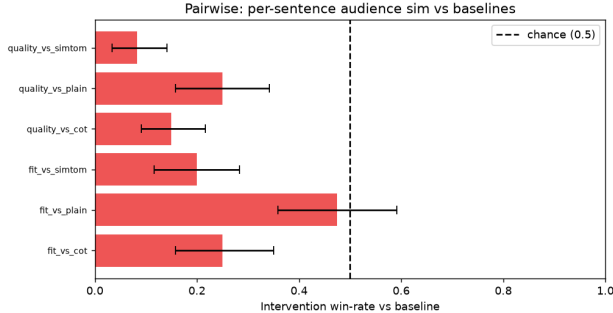


Figure 2: Pairwise win-rates for PER-SENTENCE against each baseline on overall quality and audience fit. The dashed line marks 0.5 (tie). Every bar falls below the line: the intervention loses on quality to all baselines and on fit to GENERIC-COT and SIMTOM.

#### 4.4 Homogenization and cost (H5)

Table 4 reports diversity and cost. PER-SENTENCE is the *most* homogenized condition (highest self-BLEU, lowest distinct-2, tied with PLAIN) and simultaneously the longest (170 words vs.  $\sim 147$ ) and most token-expensive (2480 thinking characters). SIMTOM is the *least* homogenized. So the intervention does not buy diversity with its extra tokens—it costs the most and produces the most generic prose. **H5 is refuted** (figure 4).

#### 4.5 Cross-family replication (LLAMA-3.3-70B)

The same ordering holds with a completely different generator. On the 48 explanatory tasks, PER-SENTENCE is worst on fit (5.52 vs. SIMTOM/GENERIC-COT 5.90) and engagement (5.02 vs. SIMTOM 5.46), and loses pairwise to *every* baseline on both quality (win-rates 0.115–0.260) and fit (win-rates 0.167–0.333), all significant. The strongest loss is again to SIMTOM on quality (win-rate 0.115,  $p < 10^{-8}$ ). The negative result is therefore not specific to one model family.

## 5 Discussion

**The direction of the intuition is right; the mechanism is wrong.** Audience modeling clearly helps: both GENERIC-COT and SIMTOM beat PLAIN, and SIMTOM—an explicit one-shot audience model—is the best condition on *every* metric, including the adaptation-gap control that specifically measures genuine audience use. So the premise that LLMs write generically partly because they model the reader weakly is supported. What fails is the specific proposed fix. Simulating the reader every sentence does not just add nothing; it actively backfires, dragging the writing below even plain generation on overall quality.

**Why does per-sentence simulation hurt?** Three mechanisms are visible in the outputs. *First, local reactivity destroys global craft.* Choosing each next sentence from the reader’s momentary reaction yields prose that is locally coherent but loses arc, voice, and payoff. In one children’s-story task, SIMTOM builds a whimsical arc with a delightful closing button (“Death’s Help Desk: Making Life Interesting Since Forever”), whereas the per-sentence version stays plot-mechanical, keeps an adult phrase (“living forever sucks”), and drifts into heavier themes (“watching everyone I loved grow old and leave”)—each next sentence locally reasonable, the whole globally off-key for a child. *Second,*

Table 3: Adaptation gap (matched-reader minus crossed-reader fit;  $n = 12$  concepts; higher means the method tailors more to its intended reader). PER-SENTENCE has the *smallest* gap, significantly below SIMTOM, which has the largest. Best in **bold**.

|                | PLAIN | GENERIC-CoT | SIMTOM      | PER-SENTENCE ( <i>ours</i> ) |
|----------------|-------|-------------|-------------|------------------------------|
| Adaptation gap | 6.08  | 6.67        | <b>7.00</b> | 6.04                         |

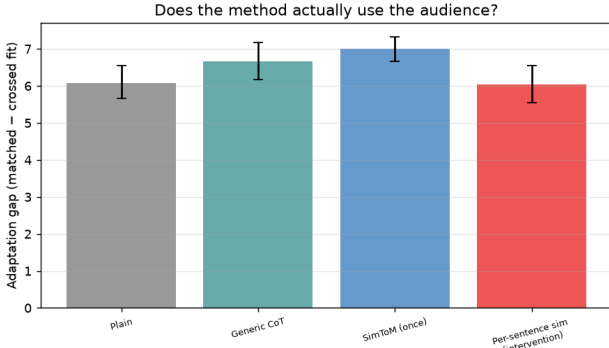


Figure 3: Adaptation gap by condition. A larger gap means the method writes differently for different readers. PER-SENTENCE adapts *least* (tied with PLAIN); SIMTOM adapts most, contradicting the hypothesis that finer-grained reader simulation yields stronger adaptation.

*it adapts less, not more* (section 4.3). A single rich audience model front-loads a coherent stance—“this is a scared 8-year-old: stay warm, concrete, gentle”—that governs the entire piece, whereas per-sentence reactions are noisier and pull the text in inconsistent directions, washing out adaptation. *Third, bloat.* The intervention runs about 15% longer and spends the most thinking tokens, yet yields worse writing—the opposite of the token-for-quality trade the hypothesis assumed.

**Granularity is the key axis.** Our result directly echoes and extends Chakrabarty et al. [2025]’s finding that more reasoning does not improve writing: even *audience-specific* reasoning fails to help when applied at fine (per-sentence) granularity, while a coarse, one-shot audience model does help. The sweet spot is “model the reader once, richly,” not “re-model every sentence.” This isolates granularity—rather than the presence or absence of audience modeling—as the design variable that decides whether audience reasoning helps or fragments the prose.

**Validity.** Several safeguards make the negative result credible. The judges are in different families from the generators, so self-preference cannot explain the ranking; pairwise judging is blind and order-balanced against position bias; only clean final text is judged, after a two-step protocol eliminated a real scaffolding-leak bug we caught mid-study (section 3); the adaptation-gap control rules out a generic-quality confound; and the full ordering replicates across two generators. The effect sizes are large (win-rates as low as 0.083) and consistent, not marginal.

**Limitations.** We test a single granularity—“every sentence.” An intermediate granularity (per-paragraph, or once-then-one-revision-pass) might recover a benefit; we did not sweep it. Fit and quality come from LLM judges rather than humans; although the judges are cross-family and the effect is large and replicated, a small human evaluation would strengthen the quality claim. The result also bounds *one* careful, faithful phrasing of the per-sentence prompt, not every conceivable one. We study short-form writing (~150 words), where global consistency is easy to maintain without per-sentence tracking; the intervention might matter more for long documents. Finally, our task set is 60 tasks (48 for replication); paired tests and bootstrap intervals mitigate this, but larger, more diverse task sets would tighten the estimates.

**Broader implications.** For practitioners, the actionable takeaway is simple and cheap: prompt the model to build one rich picture of the reader before writing, and do not ask it to re-simulate the reader sentence by sentence. More broadly, the result is a caution against importing scaffolds that

Table 4: Homogenization and cost by condition ( $\uparrow/\downarrow$  = higher/lower is better). PER-SENTENCE is the most homogenized (tied with PLAIN) *and* the longest and most token-expensive; SIMTOM is the least homogenized. Best in **bold**.

| Condition                    | distinct-2 $\uparrow$ | self-BLEU $\downarrow$ | words | thinking chars |
|------------------------------|-----------------------|------------------------|-------|----------------|
| PLAIN                        | 0.823                 | 0.406                  | 148   | 0              |
| GENERIC-CoT                  | 0.842                 | 0.383                  | 144   | 2086           |
| SIMTOM                       | <b>0.857</b>          | <b>0.363</b>           | 147   | 1762           |
| PER-SENTENCE ( <i>ours</i> ) | 0.823                 | 0.402                  | 170   | 2480           |

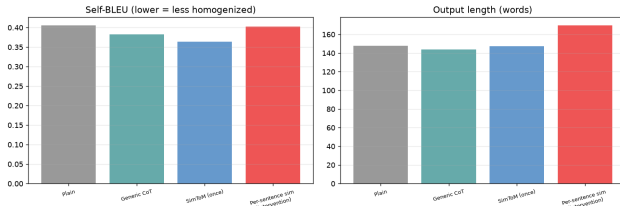


Figure 4: Diversity and cost. PER-SENTENCE combines the worst diversity (tied with PLAIN) with the highest word count and thinking-token cost, while SIMTOM achieves the best diversity at lower cost.

help *comprehension* (where per-sentence perspective tracking is known to help [Anonymous, 2024]) directly into *generation*, where global structure is part of the quality being measured.

## 6 Conclusion

We gave the first direct test of a natural, appealing idea: that an LLM writes better if it simulates its reader in chain-of-thought roughly once per sentence. Across 60 audience-targeted tasks, four conditions, two cross-family judges, an adaptation-gap control, and a replication on a second generator, the answer is a clear *no*. Per-sentence audience simulation makes writing *worse* on quality, engagement, audience fit, adaptation, and diversity, while costing the most tokens: it loses pairwise on overall quality to plain generation, generic reasoning, and one-shot audience modeling alike (win-rates 0.083–0.250, all  $p < 0.001$ ).

**Key takeaway.** The deeper intuition survives: audience modeling does help writing, but its effective form is a *single, rich audience model written once up front* (SIMTOM), which beats every alternative on every metric we measured, at lower cost. The lesson is about granularity—think about the reader once, deeply, not reactively sentence by sentence. Re-modeling the reader every sentence makes the model locally reactive and strips away the global voice, arc, and craft that make prose good.

**Future work.** Four directions follow naturally. (1) A *granularity sweep*—once, per-paragraph, per-sentence—to map where audience reasoning stops helping and starts fragmenting the prose. (2) SIMTOM *plus one revision pass* (“re-read as the reader, fix what jars”), likely the practical sweet spot, pairing the winning one-shot model with a single global critique. (3) *Human evaluation* on the SIMTOM-vs-PLAIN and PER-SENTENCE-vs-SIMTOM contrasts. (4) *Long-form* documents, where the global-consistency costs of per-sentence tracking may differ. An open question is whether per-sentence simulation’s failure is fundamental—local optimization losing global structure—or an artifact of doing it *before* writing rather than as *revision*; direction (2) would distinguish these.

**Reproducibility.** All 3,876 cached generations and judgments, the four figures, and the summary statistics are released so the full pipeline can be re-run for free.

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